Learn X+ Python Studio

Unleash Your Code!

Lesson 1B : Anaconda & Conda

ANACONDA.

CONDA

Outline :

A) What is Anaconda

B) What is Conda?

C) Live Conda Configuration

A) What is Anaconda

https://kids.britannica.com/kids/article/Austin/345848

History

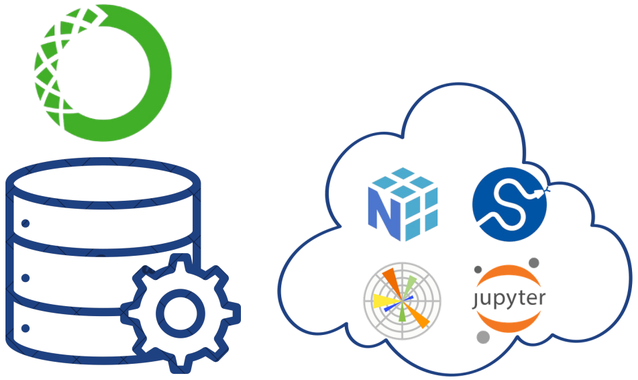
- Developed by Anaconda Inc. in 2012.
- Is used to **develop and manage** data science and AI **projects.**
- 45 million users.

Created in Austin, Texas, USA

Function

= PyPI

- Open-source **data-science** and **AI distribution** platform.
- Same function as PyPI.

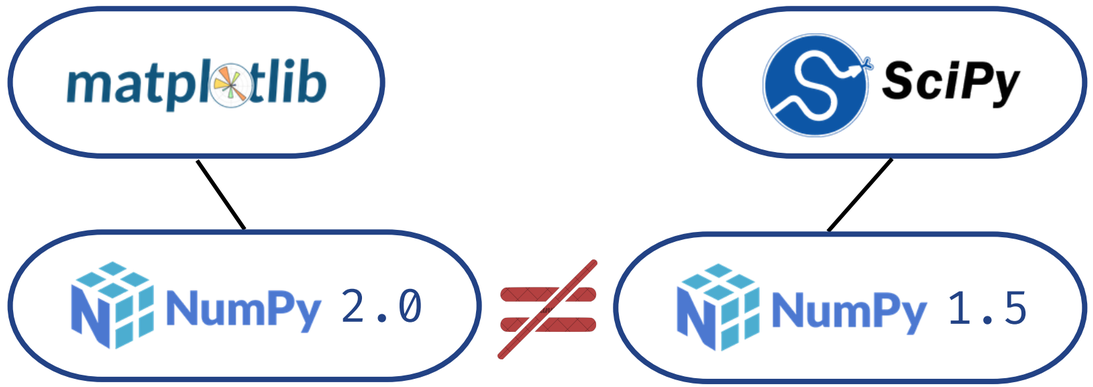


anaconda repository

A) What is Anaconda

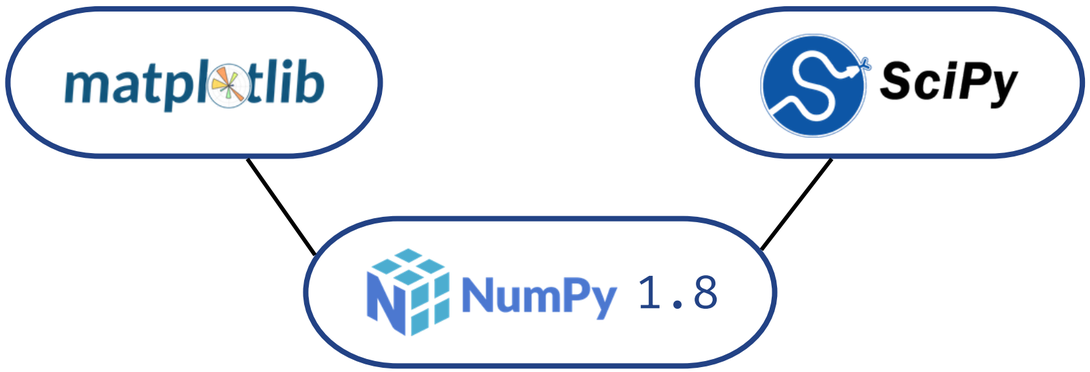
Advantages

- Anaconda **tests, builds & audit** packages.
- Ensure compatibility & security.
- Everything works right-out-the-box.



not compatible

PyPI



finds compatible version

ANACONDA.

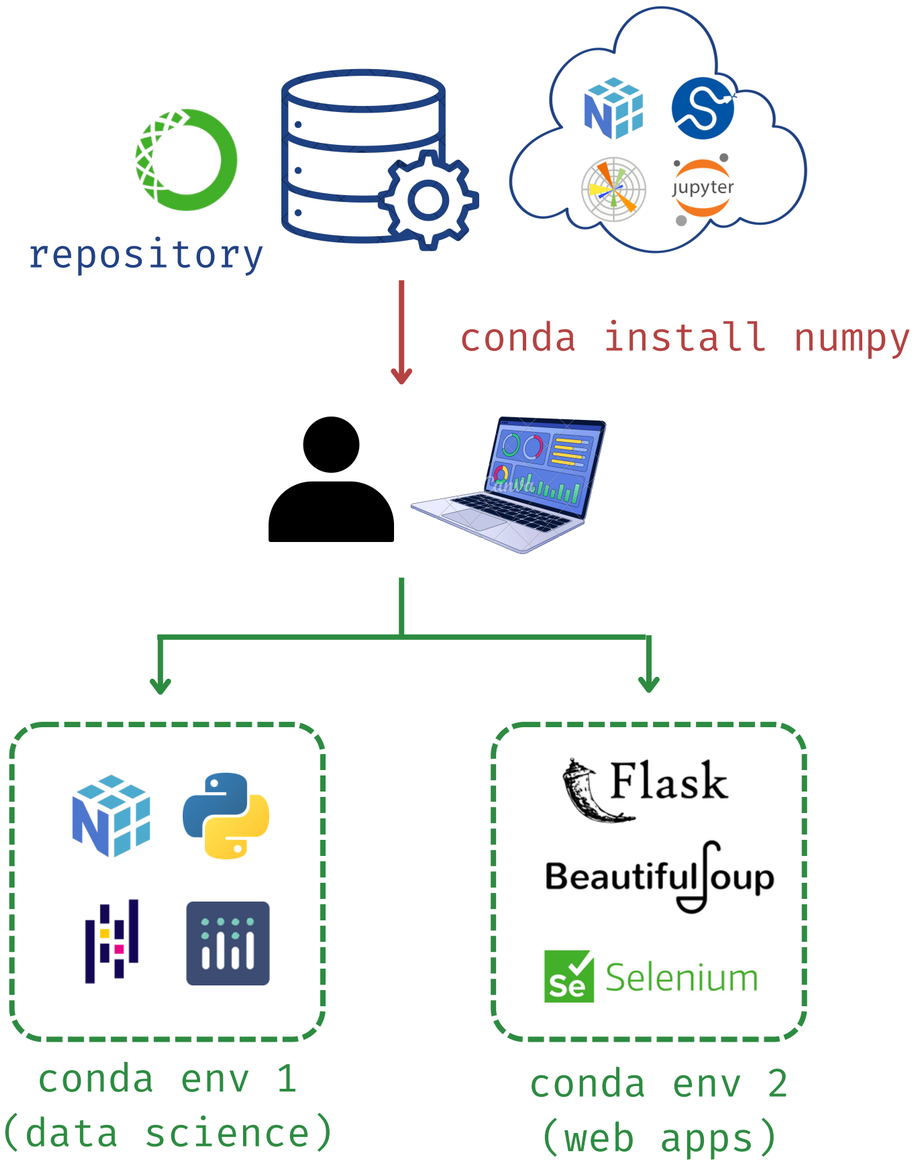
B) What is Conda?

Function

- Open-source project by Anaconda Inc.
- Both a **package manager** & **environment management system.**

= PIP and venv.

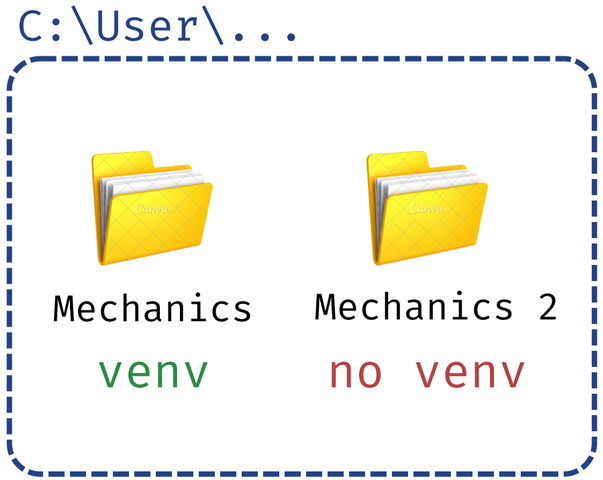
- **More flexible** than PIP & venv.



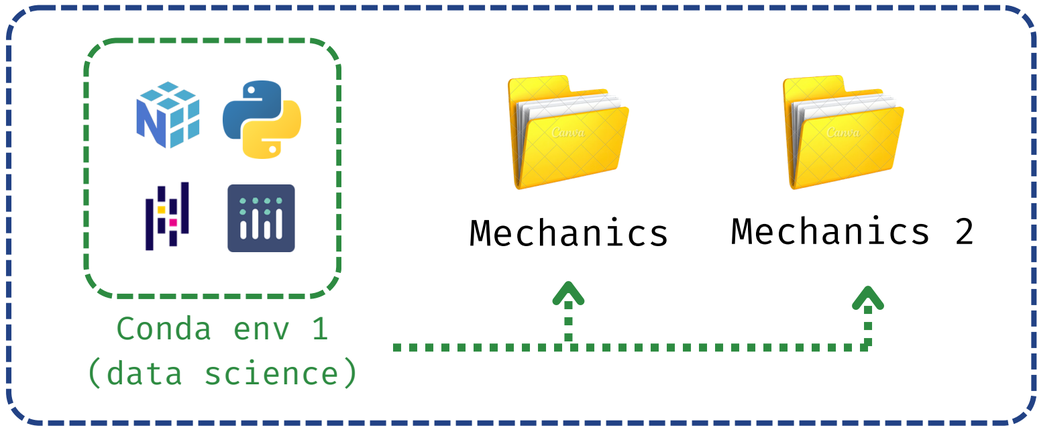
B) What is Conda?

Advantages

- Easier to use.
- Package stability.
- **Conda environments are much more flexible** than venv.



Reinstall required
for similar projects

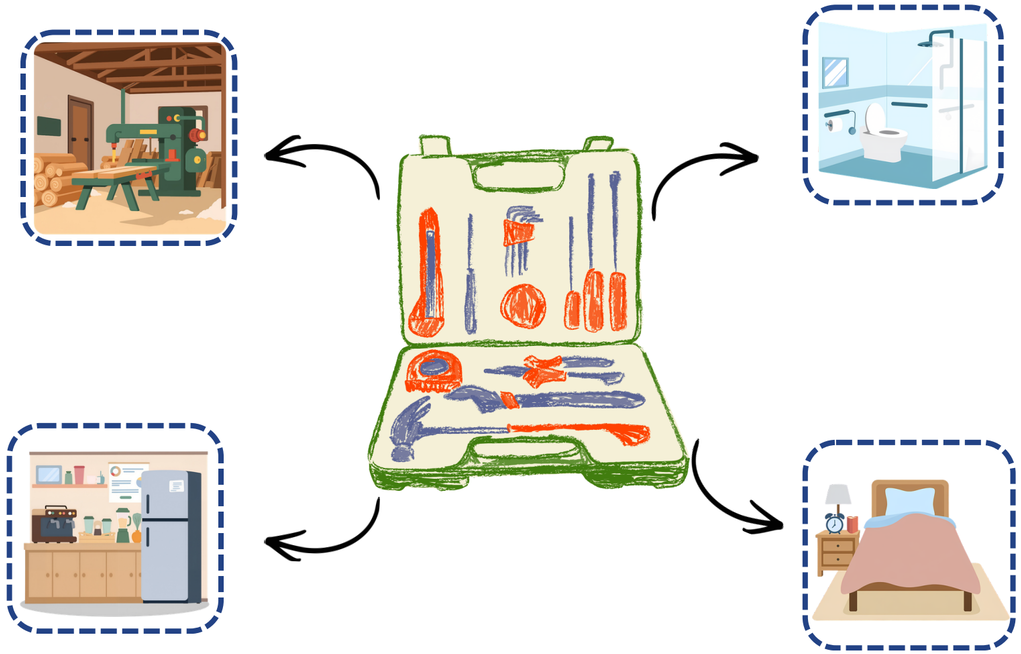


A single conda environment can be
used for many projects

B) What is Conda?

Analogy

- Venv are rooms with specific purposes.
- A person must change rooms to use the appliances there.



- Conda environments are toolboxes.
- A person can bring tools to any room and still carry out it’s function.

B) What is Conda?

Conda YAML file

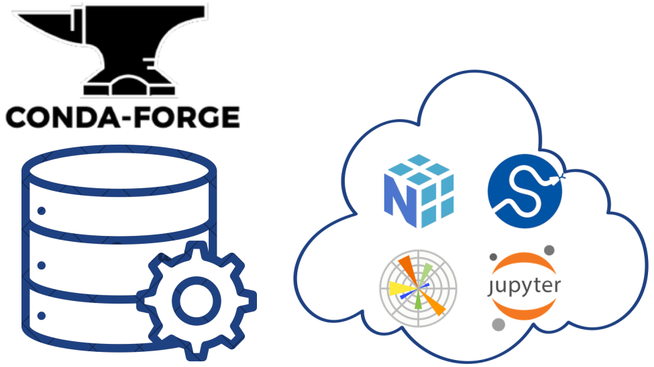
- **Easily download** project dependencies.
- **Cross-compatible** with different OS and architecture (unlike PIP).

```yaml
name: my_project_env
channels:
  - conda-forge
  - defaults
dependencies:
  - python=3.10
  - numpy=1.24
  - pandas=2.0
  - matplotlib
  - scikit-learn-
  - pip:
    - beautifulsoup4
    - requests
```

Example environment.yml file

Channels

- Repository conda downloads from.
- **Can be changed** to conda-forge.
- Community maintained:
  - More up-to-date.
  - Less secure.



conda-forge repository

C) Live Conda Configuration

Outline :

A) What is Anaconda

B) What is Conda?

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End of Lesson 1B